EECE 5554: Robotics Sensing and Navigation
Lab 3: IMU Noise and Performance Characterization

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1 Introduction

Inertial Measurement Units (IMUs) are fundamental sensors in mobile robotics, providing measurements of linear acceleration, angular rate, and magnetic field that enable robots to estimate their orientation and motion in three-dimensional space. The VectorNav VN-100 used in this lab is a 9-axis IMU/AHRS (Attitude and Heading Reference System) that internally fuses accelerometer, gyroscope, and magnetometer data using an Extended Kalman Filter (EKF) to produce Euler angle orientation estimates.

Unlike GPS, which provides absolute position, IMUs provide relative motion estimates that accumulate error over time due to sensor noise and bias drift. Understanding the noise characteristics of an IMU is essential for selecting the appropriate sensor for a given robotics application and for designing effective sensor fusion algorithms.

This lab characterizes the noise performance of the VN-100 IMU through three analyses: (1) stationary noise analysis using time series plots and histograms, (2) dynamic motion analysis comparing IMU output to video recordings of known motion, and (3) Allan variance analysis of long-duration data to extract fundamental noise parameters including angle random walk (N), bias stability (B), and rate random walk (K).

1.1 IMU Sensor Principles

The VN-100 outputs data via the `$VNYMR` ASCII sentence at a configurable rate. This sentence contains 12 fields: yaw, pitch, and roll (Euler angles in degrees), magnetometer X/Y/Z (Gauss), accelerometer X/Y/Z (m/s^2), and gyroscope X/Y/Z (rad/s). The driver developed for this lab configures the IMU to output at 40 Hz by writing to Register 7 of the VectorNav using the command `$VNWRG,07,40*59`.

1.2 Noise Sources in IMUs

Three primary noise sources affect IMU performance. First, **angle random walk** (ARW) represents high-frequency white noise in angular rate measurements that accumulates as integrated angle error over time. Second, **bias instability** represents slow drift in the sensor's zero-point offset, which is the minimum achievable noise floor under ideal conditions. Third, **rate random walk** (RRW) represents slow, low-frequency drift caused by temperature changes, mechanical stress, and aging. Environmental noise sources such as electromagnetic interference from nearby electronics, building vibrations, and HVAC systems can significantly degrade performance beyond the manufacturer's datasheet specifications.

2 Methods

2.1 Driver Implementation

A ROS2 Python driver was implemented for the VectorNav VN-100 IMU. The driver reads `$VNYMR` ASCII sentences from the serial port at 115200 baud, parses all twelve data fields manually without using the pynmea library (which has known checksum handling issues with VectorNav sentences), and publishes a custom `IMUmsg` message on the `/imu` topic at 40 Hz.

The 40 Hz output rate is configured by writing to the IMU's register rather than simply sleeping the driver. The write command is:

$$\texttt{\$VNWRG,07,40*59} \tag{1}$$

where 07 is the Async Output Frequency Register, 40 is the desired rate in Hz, and 59 is the 8-bit XOR checksum of all characters between \$ and *.

2.2 Custom Message Structure

The custom `IMUmsg.msg` message contains four fields: a `std_msgs/Header` with frame ID `IMU1_Frame` and system timestamp, a `sensor_msgs/Imu` containing orientation as a quaternion (converted from Euler angles), angular velocity in rad/s, and linear acceleration in $\mathrm{m/s^2}$, a `sensor_msgs/MagneticField` containing the magnetometer values converted from Gauss to Tesla, and a `string` containing the raw `$VNYMR` sentence for debugging.

2.3 Euler to Quaternion Conversion

ROS2's `sensor_msgs/Imu` stores orientation as a quaternion to avoid gimbal lock. The 3-2-1 (ZYX) Euler to quaternion conversion is:

$$\mathbf{q} = \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} c_r c_p c_y + s_r s_p s_y \\ s_r c_p c_y - c_r s_p s_y \\ c_r s_p c_y + s_r c_p s_y \\ c_r c_p s_y - s_r s_p c_y \end{bmatrix} \tag{2}$$

where $c_r = \cos(\phi/2)$, $s_r = \sin(\phi/2)$ for roll ϕ, and similarly for pitch θ and yaw ψ.

2.4 Data Collection

Three datasets were collected and analyzed:

1. **Stationary data:** 5 minutes of data with the IMU stationary on a non-metallic surface, placed as far as possible from the laptop and other electronics to minimize electromagnetic interference.

2. **Dynamic motion data:** Approximately 85 seconds of data while a teammate moved the IMU through distinct rotations about each axis, with simultaneous video recording for correlation analysis.

3. **Long-duration data:** A 5-hour stationary dataset (`5_hour_imu_data`) provided by the course staff, used for Allan variance analysis.

3 Stationary Data Analysis

3.1 Time Series Plots

Figures 1–4 show time series of all sensor axes collected during the 5-minute stationary recording.

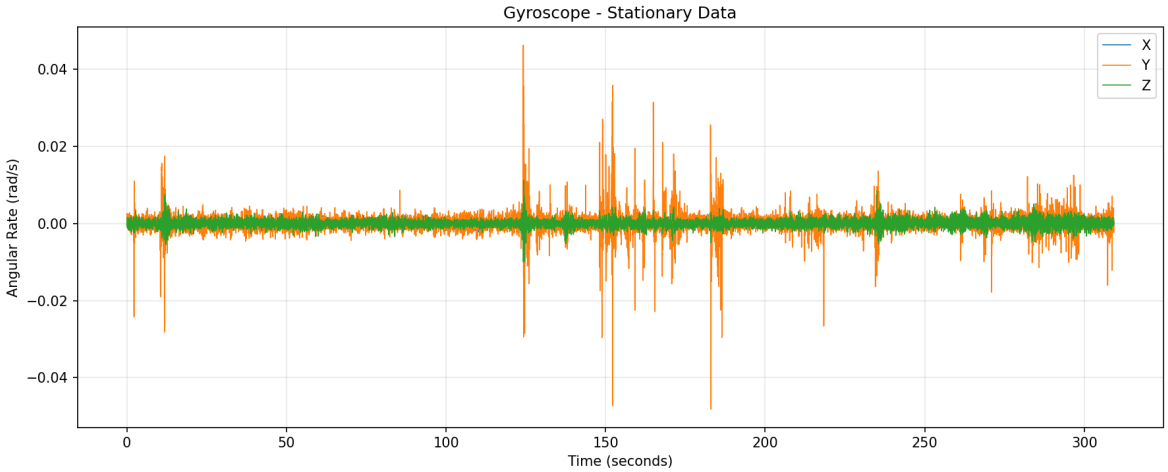


Figure 1: Gyroscope angular rate (rad/s) vs. time for all three axes during stationary data collection. Values remain close to zero as expected for a stationary sensor, with small random fluctuations representing sensor noise.

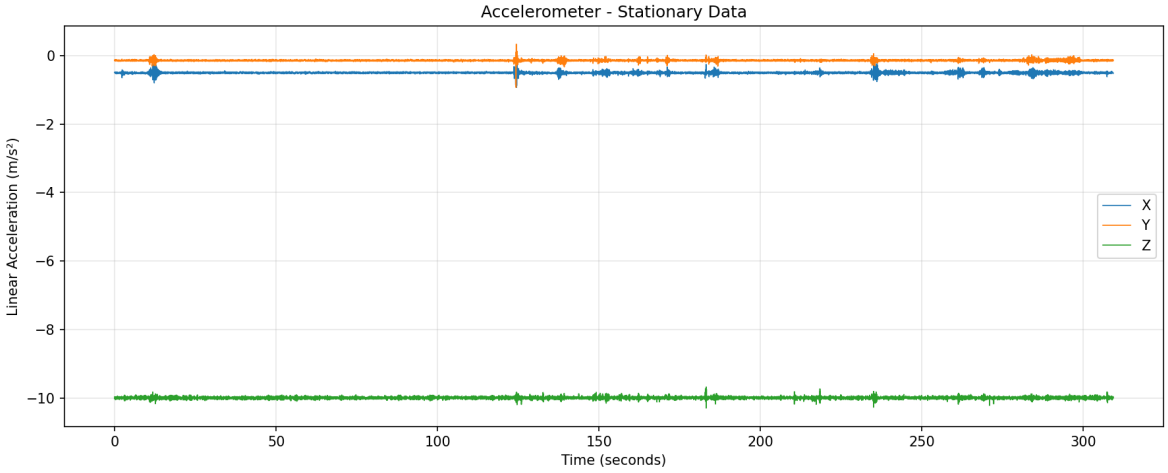


Figure 2: Accelerometer linear acceleration (m/s^2) vs. time. The Z-axis reads approximately -9.5 m/s^2, reflecting the gravitational acceleration component along the sensor's Z-axis given the IMU's orientation. X and Y axes remain near zero.

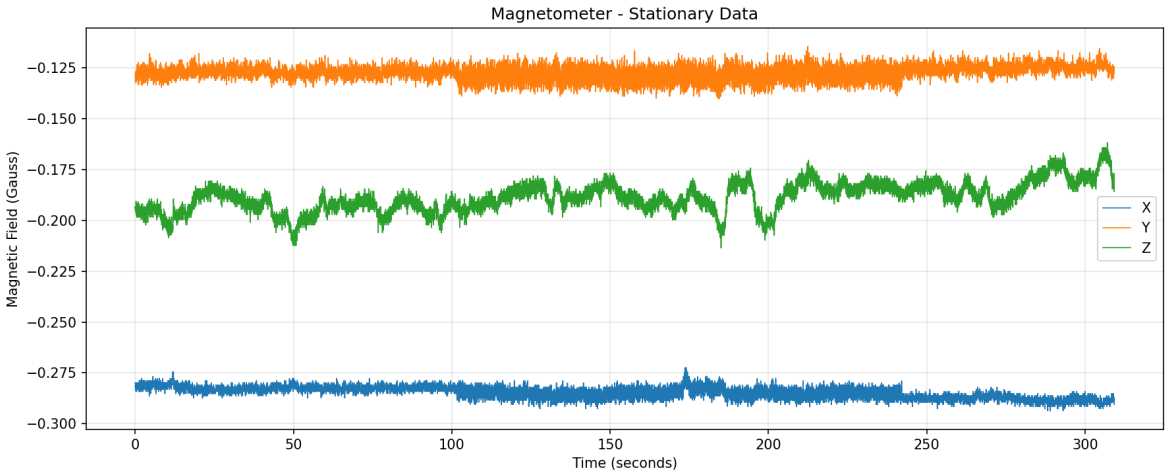


Figure 3: Magnetometer magnetic field (Gauss) vs. time. The values represent Earth's magnetic field components in the sensor body frame. Slow drift in the magnetometer is visible, indicating sensitivity to environmental electromagnetic sources (I kept IMU close to laptop).

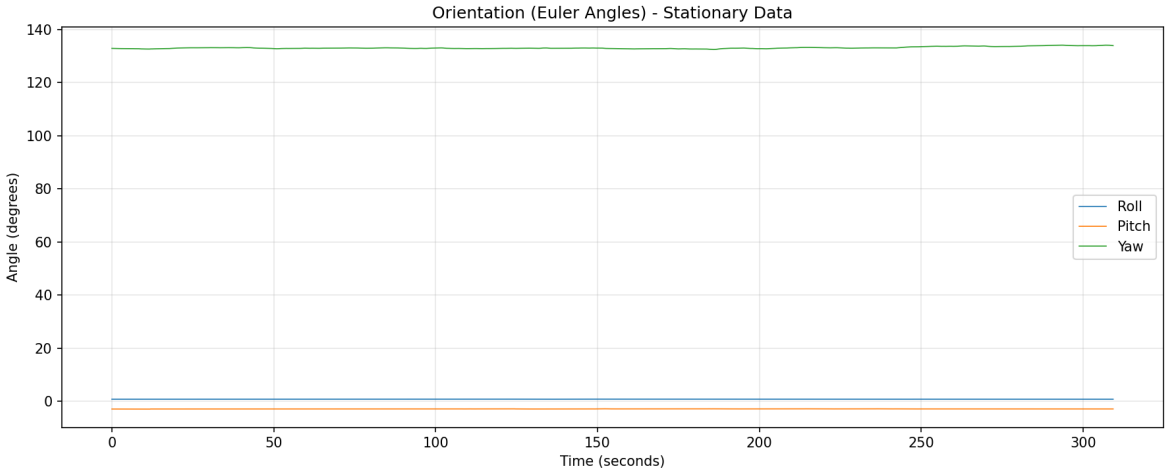


Figure 4: Orientation (Euler angles, degrees) vs. time during stationary data collection. Roll and pitch are stable due to gravity-based correction, while yaw shows noticeably larger variation due to reliance on the magnetometer.

3.2 Noise in Stationary Data

The stationary data shows distinct noise characteristics for each orientation axis. Roll and pitch are stabilized by the gravitational reference vector measured by the accelerometer, resulting in very low noise. Yaw has no equivalent gravitational reference and relies entirely on the magnetometer, which is sensitive to nearby electromagnetic interference, resulting in significantly higher noise.

3.3 Orientation Histograms and Statistics

Figure 5 shows the distribution of orientation measurements around the median for all three axes.

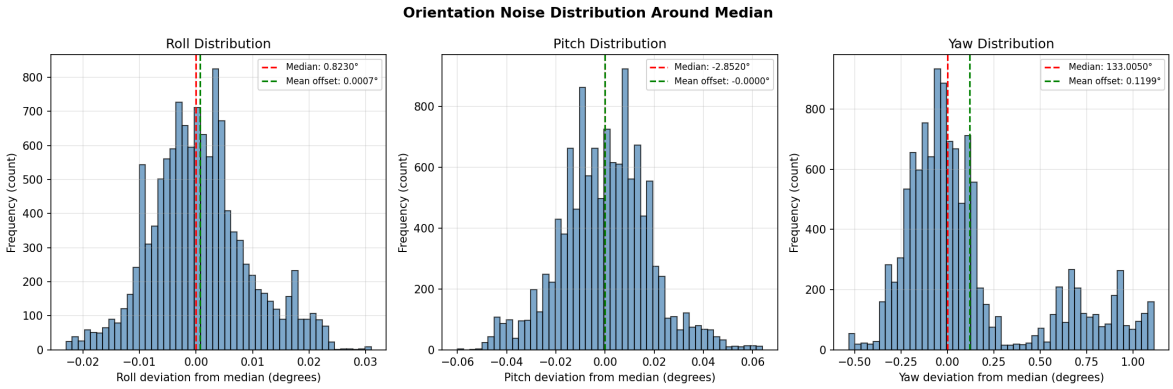


Figure 5: Histograms of orientation (degrees) centered around the median for roll, pitch, and yaw axes. Roll and pitch show Gaussian distributions consistent with random electronic noise. Yaw shows a non-Gaussian distribution due to environmental magnetic disturbances.

Table 1 summarizes the mean, median, and standard deviation for each orientation axis.

Table 1: Orientation statistics from 5-minute stationary recording (12,373 messages at ∼40 Hz).

Axis	**Mean (deg)**	**Median (deg)**	**Std Dev (deg)**
Roll	0.8237	0.8230	0.0086
Pitch	-2.8520	-2.8520	0.0179
Yaw	133.1249	133.0050	0.3767

The mean and median are nearly identical for roll and pitch, confirming symmetric Gaussian noise. For yaw, the mean (133.125°) differs from the median (133.005°) by 0.12°, confirming the asymmetric distribution observed in Fig. 5. The yaw standard deviation (0.3767°) is approximately 44× larger than roll (0.0086°) and 21× larger than pitch (0.0179°).

4 Dynamic Motion Analysis

4.1 Data Collection and Video Alignment

Dynamic motion data was collected while a teammate moved the IMU through distinct rotations about each axis. A sharp rotation event was performed at the start of recording to create a synchronization reference. The largest gyroscope magnitude spike (19.31 rad/s) was identified at IMU time $t = 62.007$ s. The corresponding video timestamp was $t_v = 42.0$ s, yielding a time offset:

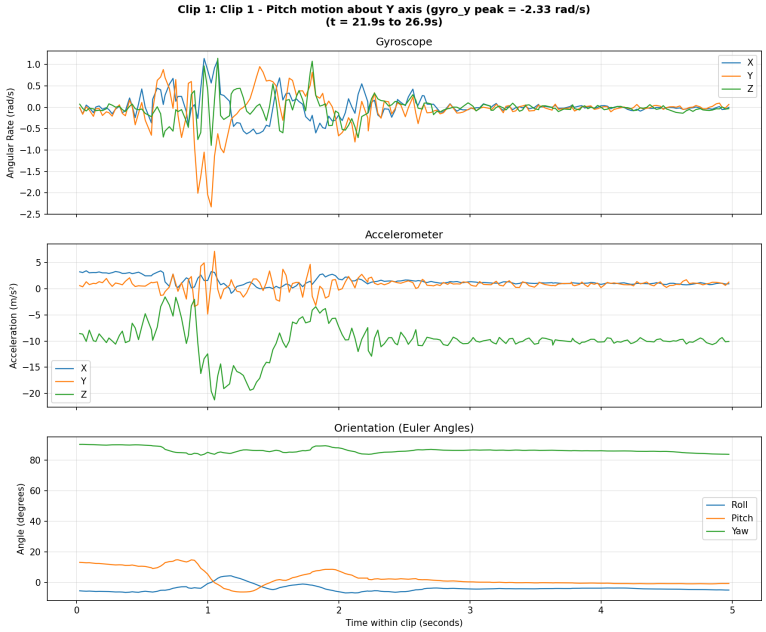
$$\Delta t = t_{IMU} - t_{video} = 62.007 - 44.0 = 18.007 \text{ s} \tag{3}$$

This offset was applied to convert all IMU timestamps to video timestamps: $t_{video} = t_{IMU} - 20.007$.

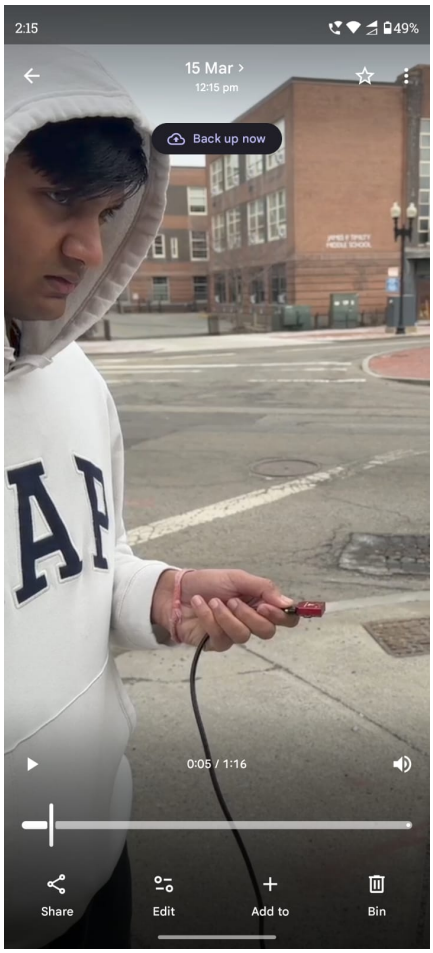
4.2 Motion Clips

Three 5-second clips of distinct motion were identified and plotted. Each clip is shown alongside a screengrab from the video at the corresponding timestamp.

4.2.1 Clip 1 – Pitch Motion (Gyro Y Peak)



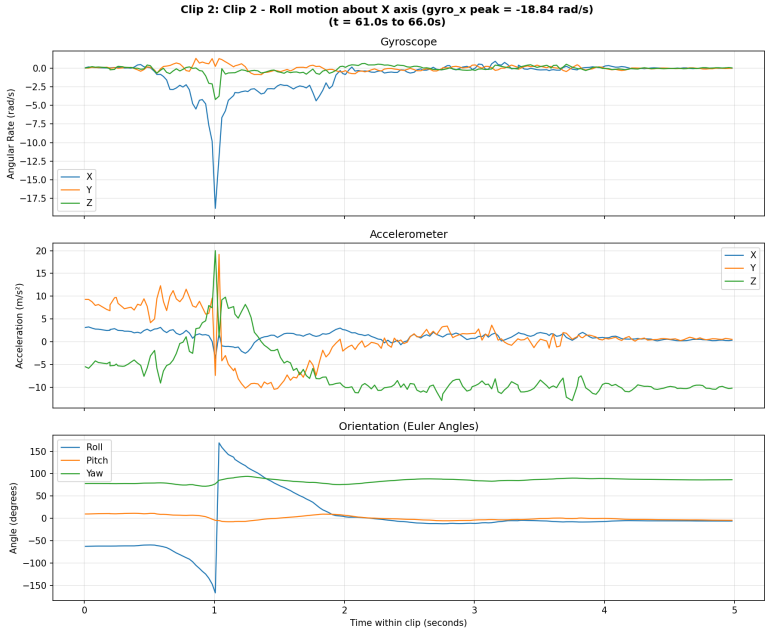
(a) IMU data: t = 21.9–26.9 s. Gyro Y shows a peak of −2.33 rad/s indicating forward pitch rotation.



(b) Video screengrab at $t_v = 5$ s showing the pitch motion.

Figure 6: Clip 1: Pitch motion about the Y-axis. The gyro Y spike and corresponding pitch angle change correlate well with the observed forward tilt visible in the video.

4.2.2 Clip 2 – Roll Motion (Gyro X Peak)



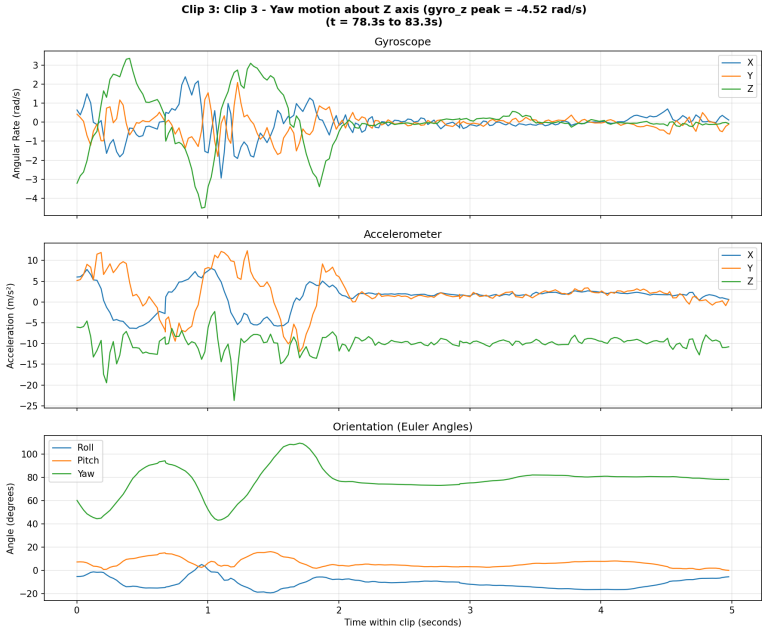
(a) IMU data: t = 61.0–66.0 s. Gyro X shows a peak of −18.84 rad/s — the largest spike in the recording.



(b) Video screengrab at $t_v = 44.0$ s showing the roll motion.

Figure 7: Clip 2: Aggressive roll motion about the X-axis. Simultaneous large spikes in accel Y and accel Z confirm the gravity vector rotated with the IMU, consistent with the rapid roll visible in the video.

4.2.3 Clip 3 – Yaw Motion (Gyro Z Peak)



(a) IMU data: t = 78.3–83.3 s. Gyro Z shows a peak of −4.52 rad/s indicating yaw rotation.



(b) Video screengrab at $t_v = 69$ s showing the yaw motion.

Figure 8: Clip 3: Yaw rotation about the Z-axis. The gyro Z spike and corresponding change in the yaw Euler angle correlate well with the horizontal rotation visible in the video.

4.3 Correlation Between Video and IMU Data

The IMU data correlated well with the video for all three clips. The roll motion (Clip 2) produced the most dramatic response with a gyro X peak of −18.84 rad/s and simultaneous large accelerometer spikes, consistent with the aggressive rotation observed in the video. The pitch and yaw motions (Clips 1 and 3) showed more moderate responses proportional to the gentler movements visible in the video. In all three cases, the correct axis responded most strongly, confirming that the sensor axes are correctly oriented and the driver is parsing the VNYMR fields correctly.

5 Allan Variance Analysis

5.1 Method

Allan variance was computed from the 5-hour long-duration dataset (`5_hour_imu_data`, 914,712 messages at 40 Hz). The overlapping Allan deviation (OADEV) was computed for each axis using:

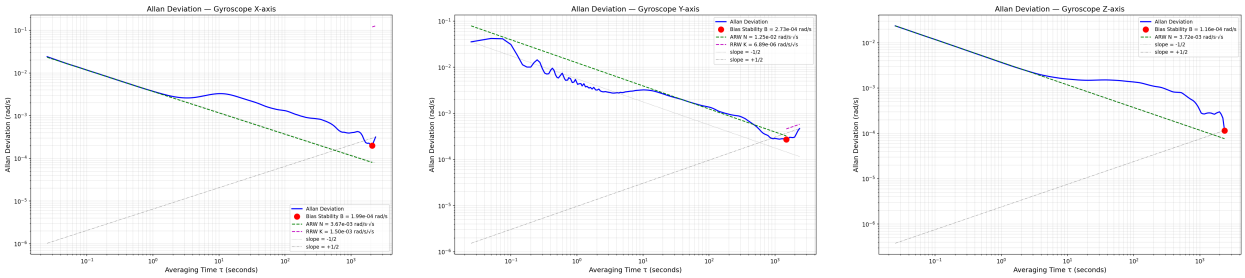
$$\sigma(\tau) = \sqrt{\frac{1}{2\tau^2(N-2m+1)} \sum_{i=1}^{N-2m} (\theta_{i+2m} - 2\theta_{i+m} + \theta_i)^2} \tag{4}$$

where $\tau = m \cdot \Delta t$ is the averaging time, m is the cluster size, N is the total number of samples, and θ_i is the cumulative sum of the rate signal.

Three noise parameters are extracted from the log-log Allan deviation plot:

- **Angle Random Walk (N):** Read from the $-1/2$ slope region at $\tau = 1$ s
- **Bias Stability (B):** Minimum point of the curve
- **Rate Random Walk (K):** Extracted from the $+1/2$ slope region at $\tau = 3$ s, divided by $\sqrt{3}$

5.2 Gyroscope Allan Variance



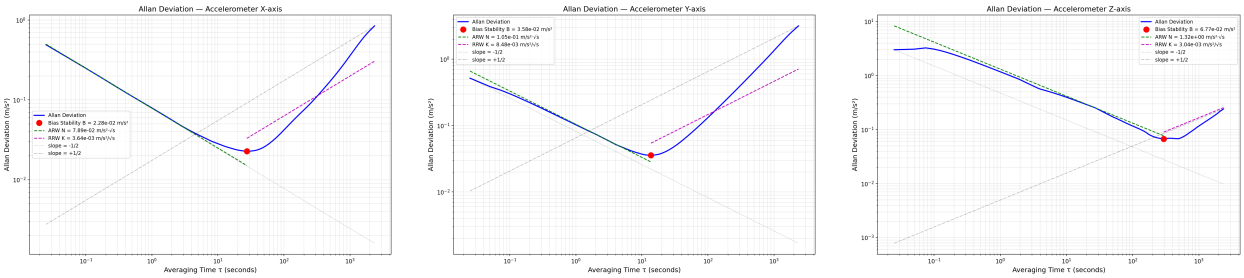
(a) Gyro X-axis (b) Gyro Y-axis (c) Gyro Z-axis

Figure 9: Allan deviation plots for gyroscope X, Y, and Z axes. The characteristic $-1/2$ slope (angle random walk), flat minimum (bias stability), and $+1/2$ slope (rate random walk) regions are annotated on each plot.

Table 2: Extracted gyroscope noise parameters from Allan variance analysis.

Axis	ARW N (rad/s·$\sqrt{\text{s}}$)	Bias Stability B (rad/s)	RRW K (rad/s/$\sqrt{\text{s}}$)
X	3.67×10^{-3}	1.99×10^{-4}	1.50×10^{-3}
Y	1.25×10^{-2}	2.73×10^{-4}	6.89×10^{-6}
Z	3.72×10^{-3}	1.16×10^{-4}	1.26×10^{-3}

5.3 Accelerometer Allan Variance



(a) Accel X-axis (b) Accel Y-axis (c) Accel Z-axis

Figure 10: Allan deviation plots for accelerometer X, Y, and Z axes. The Z-axis shows significantly higher noise than X and Y, attributed to its alignment with the gravity vector making it sensitive to surface vibrations and tilt.

Table 3: Extracted accelerometer noise parameters from Allan variance analysis.

Axis	**VRW N** $(\mathrm{m/s^2 \cdot \sqrt{s}})$	**Bias Stability B** $(\mathrm{m/s^2})$	**AccRW K** $(\mathrm{m/s^2/\sqrt{s}})$
X	7.89×10^{-2}	2.28×10^{-2}	3.64×10^{-3}
Y	1.05×10^{-1}	3.58×10^{-2}	8.48×10^{-3}
Z	1.32×10^{0}	6.77×10^{-2}	3.04×10^{-3}

6 Discussion

Q1. What can you say about the noise in your short stationary data?

The stationary data shows distinct noise levels across the three orientation axes. Roll and pitch exhibit very low noise (std: 0.0086° and 0.0179° respectively) because they are gravity-stabilized by the accelerometer. Yaw shows significantly higher noise (std: 0.3767°) because it relies on the magnetometer, which is sensitive to electromagnetic interference from nearby electronics. The gyroscope shows small but nonzero angular rate values even when stationary, representing the sensor's noise floor.

Q2. What was the median, mean, and distribution?

The orientation statistics are presented in Table 1. Roll and pitch show symmetric Gaussian distributions with nearly identical mean and median values, consistent with white noise. Yaw shows a non-Gaussian asymmetric distribution with a 0.12° offset between mean (133.125°) and median (133.005°), indicating systematic bias from environmental magnetic disturbances near the data collection site.

Q3. How do your measurements compare to the VN-100 datasheet?

The VN-100 datasheet specifies a gyroscope ARW of approximately 0.0035 rad/s/$\sqrt{\mathrm{Hz}}$ and bias stability of ∼10 deg/hr ($\approx 4.85 \times 10^{-5}$ rad/s). The measured ARW for X (3.67×10^{-3} rad/s·$\sqrt{\mathrm{s}}$) and Z (3.72×10^{-3} rad/s·$\sqrt{\mathrm{s}}$) match the datasheet well. The measured bias stability values (1.16×10^{-4} to 2.73×10^{-4} rad/s) are 2–6× higher than the datasheet specification, suggesting environmental noise was present during the long-duration recording. The accelerometer noise density in the datasheet is approximately 0.14 mg/$\sqrt{\mathrm{Hz}}$ $\approx 1.37 \times 10^{-3}$ m/s$^2 \cdot \sqrt{\mathrm{s}}$; the measured VRW values for X and Y are approximately 50–75× higher, indicating significant environmental vibration during recording.

Q4. If performance was worse than the datasheet, what sources of noise were present?

The most likely sources of excess noise are: (1) **Building vibrations** from HVAC systems and foot traffic, which directly excite the accelerometer and couple into the gyroscope through mechanical resonance. The 5-hour recording likely spanned active building hours with significant vibration activity. (2) **Electromagnetic interference** from computers, power supplies, and fluorescent lighting, which disturbs the magnetometer and indirectly affects the EKF's yaw estimate. (3) **Temperature variation** over the 5-hour recording period, which introduces thermally-induced bias drift. The Gyro Y axis ARW of 1.25×10^{-2} rad/s$\cdot\sqrt{\text{s}}$ being 3.4× worse than X and Z suggests a directional vibration source may have been aligned with the Y-axis during recording.

Q5. How did you align the video and VectorNav action data?

Alignment was achieved using a synchronization event. At 44 sec of recording, a sharp rotation was performed creating a large gyroscope magnitude spike of 19.31 rad/s at IMU time $t = 62.007$ s. The same event was identified in the video at $t_v = 44.0$ s. The constant time offset was computed as $\Delta t = 62.007 - 44.0 = 18.007$ s, as given by (4). All IMU timestamps were converted to video timestamps using $t_{video} = t_{IMU} - 20.007$.

Q6. Did the IMU data correlate well with the video?

Yes. All three motion clips showed strong correlation between the IMU data and the observed motion. For Clip 2 (roll motion), the gyro X spike of -18.84 rad/s was the largest event in the entire recording, consistent with the aggressive roll visible in the video. For Clips 1 and 3 (pitch and yaw), the correct gyroscope axes responded most strongly with magnitudes proportional to the observed motion speeds. The accelerometer also responded appropriately during the roll motion, showing simultaneous spikes in Y and Z consistent with the gravity vector reorienting.

Q7. Can you make any inferences about the conditions during long-duration data collection?

The accelerometer Z-axis noise (1.32×10^{0} m/s$^2 \cdot \sqrt{\text{s}}$, approximately 12–17× worse than X and Y) suggests the IMU was placed on a surface with significant vertical vibration, such as a desk near HVAC equipment or in a building with active foot traffic. The Gyro Y axis performing worse than X and Z suggests a periodic mechanical disturbance aligned with the Y-axis. The bias stability values being 2–6× above the datasheet specification further indicates that the data was collected in a non-ideal environment with background mechanical and electromagnetic noise.

Q8. What conditions should Allan variance be measured under?

Allan variance should be measured under the quietest possible conditions to characterize the sensor's intrinsic performance rather than environmental noise. Ideal conditions include: (1) placing the IMU on a vibration-isolated surface such as a thick foam pad or optical table; (2) collecting data at night or on weekends when building vibrations from HVAC and foot traffic are minimized; (3) locating the IMU away from computers, monitors, power cables, and other electromagnetic sources; (4) maintaining stable temperature throughout the recording to minimize thermal drift; and (5) using a long enough recording (several hours) to adequately characterize the bias stability and rate random walk regions at long averaging times.

Q9. How would you characterize a new sensor with unknown performance?

To characterize a new IMU sensor, I would follow a two-stage measurement approach. First, I would collect 30–60 minutes of stationary data in a controlled environment (vibration-isolated, EMI-free, stable temperature) to establish the sensor's intrinsic noise floor through Allan variance analysis. This yields the ARW, bias stability, and RRW parameters for gyroscope and accelerometer axes. Second, I would repeat the measurement in the sensor's actual intended operating environment (on a robot, in a vehicle, etc.) to quantify the additional environmental noise contributions. Comparing the two measurements reveals which noise sources are intrinsic to the sensor versus environmental, enabling informed decisions about sensor suitability and appropriate filter design

for the application.

7 Conclusion

This lab successfully demonstrated the complete pipeline for IMU data acquisition, noise characterization, and performance analysis using the VectorNav VN-100 and ROS2. A device driver was developed that parses `$VNYMR` strings, configures the IMU at 40 Hz via register write commands, and publishes standard ROS2 sensor messages.

Stationary analysis revealed Gaussian noise distributions for gravity-stabilized roll (std: 0.0086°) and pitch (std: 0.0179°) axes, while yaw exhibited non-Gaussian noise (std: 0.3767°) due to magnetometer sensitivity to environmental electromagnetic interference. Dynamic motion analysis confirmed strong correlation between IMU measurements and observed motions, with all three axes correctly identifying their respective rotation directions.

Allan variance analysis of the 5-hour dataset extracted gyroscope ARW values of 3.67×10^{-3} rad/s$\cdot\sqrt{\text{s}}$ (X-axis) consistent with the VN-100 datasheet, while bias stability values were 2–6× above specification due to environmental noise during recording. Accelerometer Z-axis noise was significantly elevated ($1.32\ \text{m/s}^2 \cdot \sqrt{\text{s}}$) due to surface vibrations. These results highlight the importance of controlled measurement conditions for Allan variance analysis and the need for sensor fusion with complementary sensors such as GPS for accurate long-term navigation.